Board A2-01 — Rev2 Bring-Up Log

Azul Main Controller Rev2 · JLC order SMT026080860985 · Batch A2 · Procedure: docs/hardware/rev2-bring-up.md

Field	Value
Serial	**A2-01**
Date started	
Date completed	
Firmware version flashed	
Photo captured (top / bottom)	
Signed off by	

Probe conventions

One GND clip, red probe moves. Set the DMM mode per row (VDC / VAC / Ω / mA).

GND clip point:

- **§ 3.5 (pre-power inspection, C5 not yet installed):** clip black lead to **U3 (LM2596) tab** — the large SMD GND pad on top of the board with the thermal-via array.
- **§ 4 onward (C5 hand-installed):** clip black lead to **C5 (–) leg** — through-hole radial cap negative leg, silkscreen "–".

Notation: `Cn(+)` / `Cn(-)` = positive/negative leg of cap *n*. `Un.k` = pin *k* of chip *n*. `J2.k` = screw *k* of terminal block. `R_JMPn.X-side` = the pad of R_JMPn on the named side (e.g., `R_JMP3.LDO-side` = the pad closer to U4). `U(4+n).1` = the zone-*n* MOC3062M pin 1 (Zone 1 → U5, Zone 12 → U16).

Special cases:

- Current measurements (§ 5E boot / idle) — DMM in-line in the 24 VAC hot lead between wall wart and J2 pin 1, mA range.
- Continuity (§ 3.5) — DMM in Ω/continuity mode with board fully de-powered, all R_JMPs OUT.

Rail voltage reference

Rail	Probe (red → black)	Mode	Target	Tolerance
J2 (24 VAC input)	J2.1 → J2.2	VAC	24.0 V	± 10 %
+VRAW (after bridge, before R_JMP2)	C5(+) → C5(–)	VDC	~30 V	± 15 %
Buck output (before R_JMP3)	C7(+) → GND clip	VDC	5.00 V	± 5 %
+5V rail (after R_JMP3)	R_JMP3.LDO-side → GND clip	VDC	5.00 V	± 5 %
LDO output pin	U4.5 → GND clip	VDC	3.30 V	± 3 %
+3V3 rail (after R_JMP4)	R_JMP4.WROOM-side → GND clip	VDC	3.30 V	± 3 %
MOC LED anode, zone *n* ON	U(4+n).1 → GND clip	VDC	~3.0 V	—
MOC LED anode, zone *n* OFF	U(4+n).1 → GND clip	VDC	0 V	—

§ 3 — Pre-power inspection

3.1 WROOM module (U1)

- ☐ All 41 castellated pads have solder fillets
- ☐ No solder bridges between adjacent pads
- ☐ Center thermal pad reflowed (no visible lift or tombstone)
- ☐ Antenna keep-out zone is bare copper, no stray solder

3.2 Triac stage — spot-check 4 zones (record: ____, ____, ____, ____)

For each spot-checked zone *n* (where *n* = 1…12), verify:

- ☐ MOC3062M opto-triac (U5…U16 = zones 1…12): pin-1 dot matches silkscreen
- ☐ BT137 main triac (Q1…Q12): tab orientation matches silkscreen
- ☐ Snubber resistor (R9/R12/…/R42, 39 Ω 2010) placed
- ☐ Snubber X2 cap (C12…C23, 10 nF) placed
- ☐ LED current-limit (R7/R10/…/R40, 120 Ω 0805) in correct position
- ☐ Gate current-limit (R8/R11/…/R41, 470 Ω 0805) in correct position

3.3 Power stage

- ☐ D2 bridge rectifier: pin-1 dot on silkscreen matches “+” AC input *(rev1 had pins 1/4 swapped — confirm rev2 is fixed)*
- ☐ LM2596 buck (U3) oriented correctly
- ☐ AP2112K-3.3 LDO (U4) oriented correctly
- ☐ SMD electrolytic C6 polarity: cathode bar matches “–” silkscreen
- ☐ SMD electrolytic C7 polarity: cathode bar matches “–” silkscreen
- ☐ F1 (500 mA PTC) present, no mechanical damage

3.4 USB / programming

- ☐ J1 (JST PH 6-pin) footprint: all 6 pads clean, no bridges
- ☐ USBLC6-2 ESD chip (U2) present and oriented per silkscreen

3.5 Continuity (multimeter, Ω mode, no power, R_JMPs OUT)

Black clip on **U3 tab** or **C6 (–) pad** (C5 not installed yet).

Let the DMM reading settle for ~10 s — the SMD electrolytics need a moment to charge before the true leakage resistance is visible.

Rail check	Probe (red → black)	Expected	Measured
+VRAW → GND	C6(+) → GND clip	> 10 kΩ	
+5V → GND	C7(+) → GND clip	**3–6 kΩ** (LM2596-5.0 internal FB divider)	
+3V3 → GND	R_JMP4.WROOM-side → GND clip	> 10 kΩ	
24VAC HOT → NEUTRAL	J2.1 → J2.2	open	

> **Why the +5V rail reads a few kΩ, not > 10 kΩ:** the LM2596S-5.0 fixed-voltage variant has an internal FB divider (≈3.05 kΩ + 1.02 kΩ = ~4 kΩ from +5V to GND) hard-wired inside the chip. This is designed-in and identical across all 5 boards — not a defect. If you see > 6 kΩ or < 3 kΩ, then investigate; a value in the 4-ish kΩ range is healthy.

> **STOP** if any short is found. Diagnose before continuing.

§ 4 — Hand-install pre-power parts

Install in this order. Do NOT install R_JMP1..R_JMP4 or J6 yet.

- ☐ **C5** — 470 μF 50 V radial (Panasonic EEU-FR1H471). **POLARITY: anode (“+” longer leg) to “+” silkscreen.** Body flush, leads clipped short.
- ☐ **D3** — 1N5822 Schottky, DO-201AD. **POLARITY: cathode band on part aligned with cathode line on silkscreen.**
- ☐ **SW1** — RESET tact switch, 6 mm. Clicks and returns.

- [ ] **SW2** — BOOT tact switch, 6 mm. Clicks and returns.
- [ ] **J1** — JST PH 6-pin vertical. Shroud opening faces where pigtail will exit.
- [ ] Confirmed R_JMP1..R_JMP4 and J6 still uninstalled.

§ 5 — Staged power-up

24 VAC disconnected during any solder work. Reconnect only to measure. GND clip on **C5 (–) leg** throughout.

Stage	Action	Rail	Probe (red → black)	Mode	Target	Measured	Notes
A	No jumpers, apply 24 VAC	J2 terminals	J2.1 → J2.2	VAC	24.0 V	____ V	
A	↑	F1 continuity	F1.1 → F1.2	Ω	0 Ω	____ Ω	
B	Install **R_JMP1**	+VRAW	C5(+) → C5(–)	VDC	~30 V	____ V	D2, C5 cool?
C	Install **R_JMP2**	Buck output	C7(+) → GND clip	VDC	5.00 V	____ V	LM2596 (U3) cool?
D	Install **R_JMP3**	+5V rail	R_JMP3.LDO-side → GND clip	VDC	5.00 V	____ V	
D	↑	LDO output pin	U4.5 → GND clip	VDC	3.30 V	____ V	
E	Install **R_JMP4**	+3V3 rail	R_JMP4.WROOM-side → GND clip	VDC	3.30 V	____ V	WROOM boots
E	↑	Boot current	DMM in-line, 24 VAC hot lead	mA	150–250 mA	____ mA	
E	↑	Idle current	(same in-line insertion)	mA	80–150 mA	____ mA	~5 s after boot

> **STOP** at any stage if the measured rail is outside spec, or if any component heats in <10 s, or if the rail collapses when the R_JMP is installed. Remove the last R_JMP and diagnose.

§ 6 — Firmware flash + serial console

- [ ] USB-C pigtail into J1, cable to laptop
- [ ] Serial device enumerates on laptop: `/dev/cu.usbmodem____________`
- [ ] Serial monitor open at 115200 baud
- [ ] Press RESET (SW1) — boot banner appears

- [ ] `pio run -t upload` from `firmware/main-controller/` completes without error
- [ ] After flash, RESET produces banner. Firmware version: ______________________
- [ ] `help` command lists all commands
- [ ] `status` command runs, prints device state

§ 7 — Zone GPIO logic test (no J6, no load)

24 VAC connected. GND clip on **C5 (–)**. DMM in **VDC** mode. Probe red on the MOC3062M pin 1 pad listed for each zone (or, equivalently, far side of the 120 Ω current-limit R). Expected: ON ≈ 3.0 V, OFF = 0 V.

Zone	GPIO	Red probe	ON ≈ 3.0 V	OFF = 0 V
1	38	U5.1		
2	14	U6.1		
3	13	U7.1		
4	12	U8.1		
5	11	U9.1		
6	10	U10.1		
7	9	U11.1		
8	8	U12.1		
9	18	U13.1		
10	17	U14.1		
11	7	U15.1		
12	6	U16.1		

§ 8 — Install J6 (zone terminal block)

> **Orientation:** wire-entry openings face **interior** of board (toward WROOM). Back of block faces the edge. Getting this backwards makes the board un-wireable.

- [ ] 7× ganged 2-pos terminals mated, aligned to 14-pos footprint
- [ ] Wire-entry facing interior — confirmed twice
- [ ] Soldered from bottom, joints shiny, no bridges

§ 9 — Zone triac firing test (24 VAC + load)

Test load: 24 VAC lamp OR 200 Ω 10 W resistor OR real solenoid between Zone N and COM.

CLI command per zone: `start N 5` (fires zone N for 5 seconds).

Zone	Triac (Q1…Q12) fires?	Snubber cool?	No bleed to N±1?	Notes
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

§ 10 — Sensors + status LED

- ☐ Rain sensor: short J3.RAIN pin to GND → `status` shows `rain: closed`; open → `rain: open`
- ☐ Flow sensor: tap J3.FLOW pin to GND repeatedly → pulse counter increments in `status`
- ☐ Status LED (D5, WS2812B on GPIO 21): cycles boot pattern; visible through enclosure cover

§ 11 — Radio + cloud integration

WiFi:

- ☐ `wifi-scan` lists bench AP
- ☐ `wifi-set <SSID> <pass>` accepted
- ☐ `wifi-status` reports CONNECTED within ~10 s
- ☐ IP address: ______________________

MQTT:

- ☐ `mqtt-set <host> <port>` accepted; broker: ______________________
- ☐ After reboot, serial shows `[MQTT] Connected` within 5 s of WiFi join
- ☐ Broker side (`mosquitto_sub -t 'azul/#' -v`) receives heartbeat for this MAC

Cloud round-trip:

- ☐ `curl http://<server>/api/devices | jq` shows this MAC online:true
- ☐ Mobile app: adopt controller, see online, fire Zone 1 remotely → load energizes
- ☐ MAC address of this board: __ __ : __ __ : __ __ : __ __ : __ __ : __ __

§ 12 — Enclosure fit (first board only if not verified for batch)

- [ ] Board seats on 4 WC-25F corner bosses without forcing
- [ ] M3 thread-forming screws seat cleanly, no cracking
- [ ] Barrel jack cutout aligns with J2 pigtail exit
- [ ] Cable gland cutout aligns with J6 wire bundle exit
- [ ] N/A — already verified on earlier board

Anomalies

Note anything unusual: unexpected values, rework required, parts substituted, hesitations.

Time	Observation

Sign-off

- [ ] All rails within spec (§ 5)
- [ ] All 12 zones pass GPIO test (§ 7) and triac firing test (§ 9)
- [ ] Sensors + status LED verified (§ 10)
- [ ] Cloud round-trip verified (§ 11)
- [ ] Board labeled with serial + firmware version + date
- [ ] Photos captured (top + bottom)
- [ ] This log page transcribed into `poc/rev2/dashboard.md` and committed

Signature / initials: ________________ **Date completed:** ______________________